

Security+ 3.1 · Infrastructure Considerations

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.1** —
Infrastructure Considerations (13:48) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Infrastructure design considerations: resiliency (surviving failures via clustering, load balancing, RAID, multi-AZ/region); cost (CAPEX for purchasing servers vs OPEX pay-per-use); responsiveness (latency, CDN proximity to users); scalability (scale out by adding machines is preferred). High availability targets like 99.9% uptime allow ~8h45m downtime per year, while 99.99% allows ~52 minutes. Elasticity enables scaling up and down based on demand. Redundancy N+1 survives one failure; 2N survives two.

High-Yield Core Concepts Tested on Exam

- 99.9% uptime = ~8h45m downtime/year; 99.99% = ~52 minutes
- Elasticity: scale up and down based on demand
- Redundancy N+1 survives 1 failure; 2N survives 2
- Multi-AZ/multi-region provides resilience against datacenter failure
- Scale out (add machines) is generally preferred over scale up

Exam Tips & Classic Scenarios

- 'ability to add/remove resources based on demand' = elasticity
- 'uptime of 99.99%' = high availability
- 'tolerate failure of an entire datacenter' = multi-AZ/multi-region

Typical Exam Scenario: A service must tolerate the failure of an entire datacenter. This requires a multi-AZ or multi-region deployment.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.1 — Infrastructure Considerations)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. What is vertical scalability (scale up)?

- A)** Add more instances
- B)** Increase resources (CPU/RAM) of the existing instance
- C)** Distribute load across regions
- D)** Implement auto scaling

2. Increasing capacity by adding MORE machines (not bigger machines) is scaling:

- A)** Horizontally (scale out)
- B)** Vertically (scale up)
- C)** Down
- D)** Manually

3. Multiple clients sharing the same physical infrastructure of a provider, isolated by software, is:

- A) Multi-tenancy**
- B) Air gap**
- C) Colocation**
- D) On-premises**

4. Automatically adjust the number of instances as demand rises/falls:

- A) Auto scaling**
- B) Load balancing**
- C) Snapshot**
- D) Failover**

5. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: '99.9% uptime = ~8h45 per year allowing downtime; 99.99% = 52 min.' What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Implement the corresponding solution: 99.9% uptime = ~8h45 per year allowing downtime; 99.99% = 52 min...**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators with generic credentials**

6. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Elasticity: scales up AND down as needed'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Replace preventive technical controls with informal terms of adherence**
- C) Implement the corresponding solution: Elasticity: scales up AND down as needed...**
- D) Allow shared access among administrators with generic credentials**

7. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Redundancy N+1 = survives 1 failure; 2N = survives 2.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Redundancy N+1 = survives 1 failure; 2N = survives 2...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'ability to add/remove resources based on demand' = elasticity'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'ability to add/remove resources based on demand' = elasticity...

9. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'uptime of 99.99%' = high availability'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'uptime of 99.99%' = high availability...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

10. Corporate scenario (Infrastructure Considerations): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Scenario: service must tolerate failure of an entire datacenter = multi-AZ / multi-region.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Scenario: 'service must tolerate failure of an entire datacenter' = multi-AZ / multi-region...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — Increase resources (CPU/RAM) of the existing instance.

✓ **Why it's correct:** The correct answer is 'Increase resources (CPU/RAM) of the existing instance'. Vertical scalability, also called scale up, consists of increasing the capacity of a single server or instance by adding more hardware resources, such as CPU, RAM, storage, or bandwidth. The goal is to make the existing machine more powerful to handle greater load. It is a process generally limited by the maximum resources a single machine can support and may require a reboot. CompTIA defines vertical scalability as increasing the capacity of an existing resource, as opposed to horizontal, which adds more nodes.

✗ **Why the others are wrong:**

- *Add more instances:* Adding more instances is horizontal scalability (scale out), which increases capacity by distributing load across multiple servers or nodes. It is the opposite of vertical, which focuses on strengthening a single instance. Therefore, it does not describe scale up.
- *Distribute load across regions:* Distributing load across regions is a geographic scalability and high availability strategy, generally associated with global load balancing and cross-region replication. It is not the concept of vertical scalability, which acts on a single server by increasing its resources.
- *Implement auto scaling:* Auto scaling is a mechanism that automatically adjusts the number of instances (horizontal scalability) based on metrics such as CPU or traffic. Although it can include scale up and scale down policies, the term 'auto scaling' typically refers to horizontal elasticity, not increasing resources of an existing instance. Therefore, it does not define vertical scalability.

2) A — Horizontally (scale out).

✓ **Why it's correct:** Scaling horizontally (scale out) means increasing a system's capacity by adding more machines or instances to the resource pool, rather than increasing the power of a single machine. In cloud and distributed architecture contexts, this approach is preferred for stateless applications and microservices because it offers greater resilience, elasticity, and fault tolerance. CompTIA Security+ SY0-701 covers scalability concepts as part of availability and resilience, and the exact definition of 'adding more machines' corresponds to scale out (horizontal).

✗ **Why the others are wrong:**

- *Vertically (scale up):* Scaling vertically (scale up) means increasing the capacity of a single machine by adding more CPU, memory, storage, or processing power. This does not involve adding more machines but making an existing machine more powerful. The question specifies 'adding MORE machines (not bigger machines)', which explicitly excludes scale up. Therefore, this alternative is the opposite of the described concept.
- *Down:* Scaling 'down' (scale down) refers to reducing capacity, either horizontally or vertically, to save resources. It is not a method of increasing capacity. The question asks about increasing capacity, so this alternative is semantically incorrect and does not apply.
- *Manually:* Scaling manually is not a type of scalability but a mode of operation. Scalability can be manual or automatic, but the question asks about the direction of scaling (horizontal vs. vertical). 'Manually' does not describe the architecture of adding machines but the management process, making it a vague and technically inaccurate answer.

3) A — Multi-tenancy.

✓ **Why it is correct:** Multi-tenancy is the model in which multiple clients (tenants) share the same physical infrastructure of a cloud provider but are logically isolated by software, such as hypervisors, containers, virtual networks, and IAM policies. Each tenant sees only its own resources,

as if in a dedicated environment. CompTIA Security+ SY0-701 highlights that multi-tenancy is one of the main security challenges in the cloud, requiring rigorous isolation to prevent data leakage between clients. The definition in the question exactly matches this concept.

✗ Why the others are wrong:

- *Air gap*: Air gap is a physical security measure that completely isolates a system from external networks, with no physical or logical connection. It is the opposite of multi-tenancy, as it implies total isolation and no infrastructure sharing. Air-gapped environments are used in very high-security scenarios, such as military or critical industrial systems, and do not apply to the description of multiple clients sharing the same infrastructure.
- *Colocation*: Colocation is a model in which the client installs its own physical servers in a third-party data center, renting space, power, and connectivity. The physical infrastructure is shared (building, power), but the servers are dedicated to each client. There is no software isolation as in multi-tenancy, because each client has its own hardware. Therefore, it does not correspond to the description of 'multiple clients sharing the same physical infrastructure isolated by software'.
- *On-premises*: On-premises refers to IT infrastructure installed and managed within the organization's own facilities, without sharing with other clients. It is the opposite of multi-tenancy, as there is no cloud provider or sharing of physical resources. The question describes a cloud scenario with multiple clients, which completely excludes the on-premises model.

4) A — Auto scaling.

✓ **Why it is correct:** Auto Scaling is a service that automatically adjusts computational capacity (number of instances) based on demand metrics such as CPU usage, network traffic, or queue length. In CompTIA Security+ (domain 3.4), it is cited as a cloud elasticity and resilience solution, allowing horizontal scaling up (increasing instances) or down (decreasing) according to load, ensuring availability and cost optimization. This exactly matches the statement: 'automatically adjust the number of instances as demand rises/falls'.

✗ Why the others are wrong:

- *Load balancing*: Load Balancing distributes incoming traffic across multiple instances to prevent overload on a single server and improve availability. It does not automatically increase or decrease the number of instances; it only directs requests. Therefore, it is not the correct answer for automatic capacity adjustment.
- *Snapshot*: A Snapshot is a point-in-time copy of a storage volume (disk) or instance, used for backup, disaster recovery, or cloning. It has no relation to dynamic instance adjustment based on demand; it is a backup operation, not scalability. Thus, it is incorrect.
- *Failover*: Failover is the process of automatically redirecting traffic to a redundant system (standby) when the primary fails, ensuring continuity. It does not adjust the number of instances based on demand; it only switches to a backup resource in case of failure. Therefore, it does not meet the statement.

5) B — Implement the corresponding solution: 99.9% uptime = ~8h45 per year allowing downtime; 99.99% = 52 min....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): 99.9% uptime = ~8h45 per year allowing downtime; 99.99% = 52 min. These availability percentages are critical for understanding SLAs and redundancy requirements.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks

compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) C — Implement the corresponding solution: Elasticity: scales up AND down as needed....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): Elasticity: scales up AND down as needed.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) A — Implement the corresponding solution: Redundancy N+1 = survives 1 failure; 2N = survives 2....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): Redundancy N+1 = survives 1 failure; 2N = survives 2. N+1 redundancy provides one additional component beyond what is needed to handle the load, allowing the system to survive a single failure. 2N redundancy duplicates the entire system, allowing it to survive two failures or a complete system outage.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) D — Implement the corresponding solution: 'ability to add/remove resources based on demand' = elasticity....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): 'ability to add/remove resources based on demand' = elasticity. Elasticity refers to the cloud's ability to automatically scale resources up or down based on demand, ensuring optimal performance and cost efficiency.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting

pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) B — Implement the corresponding solution: 'uptime of 99.99%' = high availability....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): 'uptime of 99.99%' = high availability. This reflects the standard definition of high availability, which aims to ensure systems are operational and accessible for the vast majority of time.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) A — Implement the corresponding solution: Scenario: 'service must tolerate failure of an entire datacenter' = multi-AZ / multi-region....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Infrastructure Considerations): Scenario: 'service must tolerate failure of an entire datacenter' = multi-AZ / multi-region. Multi-AZ and multi-region deployments provide high availability and fault tolerance by distributing workloads across independent availability zones or geographic regions, ensuring that the failure of a single datacenter does not cause service outage.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

